# Engineering reproducibility guide, version 2

## Safe PyCharm entry point

Open `C:/Users/Alban/Documents/TPWRS_PaperB_Engineering_Validation_20261018_v2` as a separate project. Use `C:/Users/Alban/anaconda3/python.exe` (Python 3.9.7, NumPy 1.26.4, pandapower 3.2.1). Run `RUN_ENGINEERING_VALIDATION.py` with no arguments. Default STAGE="PREFLIGHT" checks consumed hashes and verifies/reuses completed preflight. Other settings: PHYSICAL_METRICS, AC_REPLAY, HEADROOM, INFORMATION_AUDIT, REPORTS, ALL. ALL executes stages in that order. Completed stage markers prevent repeat solves; per-case AC JSON and per-policy physical checkpoints preserve completed work. An unresolved prerequisite raises an explicit error. No original RUN.py or protected module is imported.

All writes are confined to this version-2 directory. Outputs use write-once atomic creation; a mismatched existing file raises an error rather than replacing evidence. Do not delete stage markers or edit a scientific configuration to force reuse under another identity. To change scientific assumptions, create a new isolated version. Execution journals are distinct from immutable scientific results.

The completed blocked version remains at `C:/Users/Alban/Documents/TPWRS_PaperB_Engineering_Validation_20261018`. The present author decision resolves its joint P/Q interpretation by explicitly declaring C_Q_new=C_P_final. It does not retroactively modify the original AC model.

## Source roots and exact identities

Protected root: `C:/Users/Alban/OneDrive - University of Groningen/Desktop/Final Paper Script`. Archive: `PAPER B/TPWRS_FINAL_ARCHIVE` below that root. Every consumed absolute path, byte size and SHA-256 appears in `07_QC_AND_INTEGRITY/CONSUMED_INPUTS.json` and the final consumed-file CSV. `SOURCE_AND_RESULT_TRACEABILITY.csv` links report findings to frozen sources and generated data. `EVALUATION_CODE_IDENTITIES.json` records isolated script identities. `AC_CONFIGURATION_V2.json` binds mappings and solver settings.

Authoritative main/report sources and PDFs are under `C:/Users/Alban/Documents/TPWRS_PaperB_Final_Signoff_20261017`; exact six identities are in `00_AUTHORITY_AND_INPUTS/AUTHORITY_IDENTITIES.csv`. Both sealed submission ZIPs and all retained Gate 1 certificate artifacts are unchanged. The earlier complete-tree inventory is a historical reference; the final check here covers every consumed file rather than rescanning unrelated files.

## Physical inputs and original code references

Physical metadata/arrays: `IMPLEMENTATION/frozen_inputs/artifacts/real_case/physical_case/FINAL_DUTCH_CASE/physical_case.json` and `physical_case_arrays.npz`. Final participant connections are `PARTICIPANT_MAPPING.json` alongside them. The balancing vector is in `IMPLEMENTATION/frozen_inputs/artifacts/real_case/reduction/REDUCTION_EFFECTS.npz`. Signed state sensitivities, offsets, rating rows and UTC ordering are loaded exactly from the frozen arrays. New exact row mapping is `00_AUTHORITY_AND_INPUTS/EXACT_ROW_MAPPING.csv`.

The native network and central-rating source absolute paths are bound by `IMPLEMENTATION/frozen_inputs/artifacts/real_case/LOCAL_SOURCE_CONFIG.json`. Reactive inputs are `IMPLEMENTATION/frozen_inputs/artifacts/ac_inputs/AC_INPUT_FREEZE.json` and `ac_inputs.npz`. Their source hashes are verified. Source PV generation P, background Q and native generator Q limits are retained unchanged.

Historical code references are `IMPLEMENTATION/source/jobs/selection.py`, `jobs/dc.py`, `jobs/ac.py`, `tpwrs/engineering.py`, `tpwrs/network.py`, `tpwrs/rolling.py`, `jobs/input_build.py`, `jobs/debit_rule_family.py` and `tpwrs/debit_rules.py`. They are inspected, not imported or executed. Native construction/solve functions are independently transcribed in the isolated `engineering_common.py` and `ac_evaluation.py`.

## Stored trajectories and chronological semantics

Annual M1-LP/M2: `RESULTS/annual_primary/year_Y/rho_0.15/annual/RULE/solution_arrays.npz`. Rolling M1/M2: `RESULTS/rolling/m1_m2/year_Y/rho_0.15/rolling/RULE/rolling_arrays.npz`. Raw U/C: `RESULTS/debit_rule_family/year_Y/UNIFORM_PREMIUM/RAW_ROLLING_POLICY.npz` and `CONSERVATIVE_STATE/RAW_ROLLING_POLICY.npz`. EC: `RESULTS/debit_rule_family/year_Y/EXPOSURE_CONTROLLED_ANNUAL_WITNESS.npz`. Y is 2029, 2030 or 2031. No action is repaired, scaled, refit or reoptimized.

The frozen card and validation decision are in `RESULTS/debit_rule_family/FITTED_CARD.json`, `EXPERIMENT_FREEZE.json` and `VALIDATION_DECISION.json`. Card parameters are fixed; rate assignments vary with realized scenario inputs. The finite-window information is not authenticated forecast-vintage information. Detailed timing and the prospective mathematical requirement are in `04_INFORMATION_TIMING/INFORMATION_STRUCTURE_AND_0700_AUDIT.md`.

## Baseline and matched AC registry

Frozen sample: `RESULTS/engineering_selection/SELECTED_POLICY_STATE_CASES.csv`, `SELECTED_HOURS.csv`, `SELECTED_STATE_REGISTRY_FREEZE.json`. Historical metrics: `RESULTS/selected_ac/CASES.csv`, `LOADING.csv`, `VOLTAGES.csv`. The predeclared sample and tolerances are in `00_AUTHORITY_AND_INPUTS/BASELINE_CASES_PREDECLARED.csv` and `BASELINE_TOLERANCES.md`. All 27 cases use the original two-map model. The subsequent full 162 historical-M2 comparisons are separately recorded.

`stage_ac.py` creates 648 common-map and 648 historical-map cases, each with four policies at the same original hour, state and PF. Original source PV/slack controls and Newton-Raphson settings are preserved: dc then flat initialization, angles, Q-limit enforcement, connectivity checking, 100 iterations and 1e-8 tolerance_mva. Failure records preserve attempts and no zero-score fallback is used.

All case JSON records are retained under `02_MATCHED_AC_VALIDATION/cases`. Aggregates include MATCHED_AC_CASES.csv, MATCHED_AC_LOADING.csv, MATCHED_AC_VOLTAGES.csv, MATCHED_AC_REACTIVE_LIMITS.csv, MATCHED_AC_PAIRWISE_COMPARISONS.csv, MATCHED_AC_PAIRWISE_SUMMARY.csv, MATCHED_AC_SUMMARY.csv and AC_MAPPING_SENSITIVITY.csv. Mapping tags must remain attached to every comparison.

## Full-year metrics, headroom and independent QC

`stage_physical.py` processes all 8,837 rows in chunks of 128 hours. It creates ANNUAL_PHYSICAL_METRICS.csv, HOURLY_MAXIMUM_EXCEEDANCE.parquet, CRITICAL_ROWS_BY_POLICY.csv and OVERLOAD_DURATION_SUMMARY.csv under `01_PHYSICAL_METRICS`. All 21 policy-year residual totals agree with stored fixed-action replay evidence. Thresholds are 1e-4 MW and the declared 1-MW reporting convention; quantiles use NumPy linear percentiles.

`stage_headroom.py` recovers background device dispatch from `PAPER B/Final/artifacts/real_case/background/background_Y.npz`, bound to its BACKGROUND_AUDIT_Y.json cache hash, and checks exact equality to the physical background. The immutable device registry preserves generator, interchange and renewable-curtailment labels. Native P/Q bounds and fixed AC PV references are inventoried. Counter-injection shares are bus-level; generator-specific adjustments are not identified. Conditional aggregate-bound results do not become a redispatch feasibility proof.

`stage_qc.py` independently aggregates all persisted AC row tables and hourly diagnostic statistics, checks native line loading from current/rating, checks alternate DC algebra on a stated sample and directly recomputes every retained critical-row flow. Existing independent full-action replay totals supply a separate objective-identity check. No extra optimizer or AC solve is hidden in QC. All scope limits are explicit in INDEPENDENT_NUMERICAL_QC.json.

## Reports, figures and manuscript changes

`build_dossier.py` reads completed data and generates five unsmoothed publication figures and report Markdown. `render_dossier.py` creates four PDFs with the bundled ReportLab runtime, and Poppler page images are visually reviewed. Figures A/B use the annual metrics CSV; C uses the hourly Parquet; D uses AC summaries; E uses the complete case registry. Figure provenance JSON gives these exact paths. Deterministic years/PF/contingencies are not treated as independent statistical trials.

Reports are the final engineering dossier, matched premium AC report, generator-headroom report and this guide. Proposed changes are `05_MANUSCRIPT_AMENDMENTS/PROPOSED_MANUSCRIPT_CHANGES.tex` and MANUSCRIPT_CHANGE_RATIONALE.md. No locked manuscript source is overwritten, and no IEEE/Zenodo publication or DOI reservation occurs.

The final verdict is conditional on retaining the declared controlled-model interpretation, adverse thermal findings and headroom/timing limits. The new reactive sensitivity never changes the original annual DC accounting certificate.